

BOOKS

Exotic Dispersals

Dispersal of Living Organisms into Aquatic Ecosystems. Edited by A. Rosenfield and R. Mann. University of Maryland, Sea Grant College Program, 112 Skinner Hall, University of Maryland, College Park, MD 20742, USA. \$35.00.

During the past decade there have been increasing activities in regard to the introductions or transfers of exotic species and their subsequent dispersal within receiving waterways. The International Council for the Exploration of the Sea (ICES) took special interest and convened working groups and subcommittees concerned with introductions in the marine environment. These interests culminated in a major symposium held in Halifax, Nova Scotia, 12–13 June 1990 entitled, 'Introductions and Transfers of Aquatic Species'. The proceedings of this symposium were published in June 1992 and have been well received within the academic and managerial communities.

Recently, a second conference was sponsored by the National Oceanic and Atmospheric Administration. The papers presented at this conference have recently been edited and compiled into a volume entitled, 'Dispersal of Living Organisms into Aquatic Ecosystems', edited by Drs. Aaron Rosenfield and Roger Mann. Published by the Maryland Sea Grant Office, the volume includes seven major chapters or subdivisions, with each having some two to six contributions authored by experts in the field. The first part of the volume (Chapters 1–4) deals with risks and impacts (effects) associated with introductions and dispersals of exotic species and a range of pathogens, parasites, predators, and unusual competitors. The second part concerns risk reduction and safety. What regional, national, and international groups have jurisdiction over introductions and subsequent dispersal of introduced species? How can the management of introduced species be carried out so that 1. unwanted introductions are avoided, 2. introductions that occur are controlled, and 3. the effects of introductions, where uncontrollable, can be mitigated?

The overall thrust of the book is of such a nature that it could be used as a textbook in any new course on introductions (it deals with freshwater and marine forms), and provide as well technical background useful to managers at the local and state level who must deal with situations resulting from introductions.

If it has a shortcoming it is that it does not have a specific chapter on effective communications such that scientists and managers can have the precise information necessary to begin to inform the general citizenry, fishermen, involved industry, and managers on how to avoid introductions in the first place, and communicate possible effects so that early, draconian measures can be taken to mitigate introductions of pathogens, parasites, pests, and unwanted species. There is little doubt that had the world seen the possible economic con-

sequences of the introduction of the Zebra mussel, more stringent measures would have been put in place well over a decade ago. Even today the public has a greater concern about the elimination of a species (biodiversity) than it does for consequences of transport of unwanted species into new zoogeographic provinces. Fortunately, this narrative by Mann, Kern, Rosenfield, and Sindermann cover the field reasonably well if not precisely.

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Green Update

Green Globe Yearbook 1992. Edited by H. O. Bergesen, M. Norderhaug & G. Parmann. Oxford University Press, produced by the Fridtjoff Nansen Institute, Norway, 300 pp.

A mass of organizations, agencies and programmes concerned with the environment have sprung up over the last 20–30 years, identifiable to those in the business by their acronyms but often indistinguishable to lesser mortals and looking like something out of a game of scrabble oversupplied with 'I's and 'U's. The United Nations Conference on Environment and Development (UNCED to the professionals, the Earth Summit to those in search of something straightforward) brought many of these institutions to a much wider audience, and hopefully showed that they have an important role to play, even if their actions are as vulnerable to the demands of politicians as those of the rest of us.

This publication has the ambitious aim to demonstrate how far environment and development problems are being solved, what the main obstacles are and what needs to be done, at the international level. It is divided into three sections. The first consists of 11 essays on different aspects of the international agenda on environmental issues. These include AIDS, the rights of indigenous peoples, population control, the wildlife trade, hazardous waste disposal, ozone depletion and climate change. A number provide an analysis and discussion of the treaties described elsewhere in the book. They provide useful reviews of these topical issues and will be of value for students and their teachers.

The second and third parts of the book are what should make this volume a particularly valuable reference work. The second part provides a brief description of each of the major international agreements, treaties and conventions on environment and development, with information on the number of contracting parties at the time of publication, any affiliated protocols and annexes, the secretariat, financing arrangements, a summary of the contents of the treaty and how it is implemented and monitored. The major marine treaties are covered, although not every regional fishery agreement. A large table summarizes parties to these conventions, but the information in this

needs to be treated with caution; it is not as accurate as the rest of the book appears to be.

The third part provides similarly condensed information on each of the major intergovernmental organizations and major international NGOs (non-governmental organizations) concerned with the environment and development (one criterion for inclusion being that they have to be concerned with both issues). Where appropriate, there is a map showing world membership; other information includes the aims of the organization, the address of the secretariat, the organizations main activities, the regular publications it produces and how it is funded.

The editors make no pretence of providing a complete overview and have included a questionnaire to be returned by readers with suggestions for improving future editions and ideas for topics for future essays. *Marine Pollution Bulletin* readers could probably usefully look at the information given on marine issues and make helpful suggestions: land-based sources of pollution and the state of the world's fisheries are two topics that spring immediately to mind.

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NEW PRODUCTS

New Diode-Array Detector for HPLC

Hewlett-Packard has just developed a new diode-array detector for liquid chromatography. The new detector is available as a stand-alone detection system or as part of an HP 1050 Series modular high performance liquid chromatography (HPLC) system.

The HP 1050 is designed for research analysis and for sophisticated routine work using spectral libraries. The new detector acquires 3-D data continuously during a peak's elution—the third spectral dimension provides the chromatographer with more data per unit time to give more reliable results.

For automated work, the new detection system features an HPLC3D ChemStation (DOS series), a PC-based controller (typically an HP Vectra PC with a 486 processor) with an HP LaserJet printer and software running under Microsoft Windows. As an example of the reliability of results from the system, the HPLC3D ChemStation software can confirm a peak's purity while quantifying the response against calibration standards.

Diode-array detectors have until now relied heavily on user interaction and were limited to research or method development work. With this new system, routine analyses such as environmental monitoring, pharmaceutical quality control and quality assurance testing of other manufactured goods can benefit from the HP 1050 Series diode-array detector's automated features such as:

- peak qualification—based on absorbance ratios of two or more simultaneously acquired signal wavelengths;
- peak purity calculations—based on comparisons of UV-Visible absorbance spectra recorded during elution;
- positive peak identification—based on spectral comparisons with archived UV-Visible absorbance spectra from spectral libraries; and
- self-validating optics and microprocessors—based on a holmium oxide filter for wavelength calibration and

an electronic system verification feature that verifies data transfer from detector to PC.

Oil Tracker

Radio Satellite Integrators Inc have developed an oil spill tracking buoy for spill trajectory modelling. The device, known as Oil Tracker is a free-floating buoy tracked via VHF or UHF radio link from either a land- or sea-based personal computer and base station. The Oil Tracker utilizes Global Position System (GPS) satellites to obtain its location. The Oil Tracker's hull, a Lagrangian-type drifter developed at MIT's Draper Laboratories and manufactured by Aanderaa Instruments, emulates the movement of spilled oil on water and therefore holds its position relative to the spill. By utilizing navigation satellites for positioning, the Oil Tracker is able to monitor the oil spill in real-time, 24 hours-a-day.

The waterproof Oil Tracker is equipped with a state-of-the-art Magellan GPS receiver module integrated with a proprietary communications controller and firmware. A VHF or UHF radio and batteries complete the package. For the base station, various real-time software interfaces are offered to track the Oil Tracker from the portable base station computer platform.

An interface to Applied Science Associates (ASA) OILMAP software package is also available. OILMAP is an interactive graphical PC-based oil spill model for world-wide application. The software provides rapid predictions of the movement of spilled oil and contains an embedded geographic information system (GIS) for mapping of response equipment, sensitive environmental areas, and oil spills.

In the unfortunate circumstance of an actual oil spill, it has the capability of monitoring the spread and movement of the oil in real-time by deploying multiple buoys. This real-time tracking system will enable co-ordination of cleanup crews in the most efficient